Making Trivial Agents Fail in $\tau/\tau^2/\tau^3$-Bench

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Claim

Zhu et al. [1] showed that τ Bench gives credit to agents that do nothing. τ^2 Bench and τ^3 Bench fixed many task quality issues pointed out in that paper, including more than 75 task fixes in τ^3 [2,3], but this specific outcome validity problem is still present.

We tested this directly on τ^3 Bench with a null agent that takes no action and says nothing. It gets 46% on airline, passing 23 of 50 tasks, and 5.3% on retail, passing 6 of 114 tasks. These are the same tasks independently identified in the open Github Issue #384 [4].

The main problem is that reward components can silently pass when there is nothing to evaluate. For instance, 44 of 50 airline tasks include a communication reward even though no communication criteria are provided. On refusal or read only tasks, this can leave only the final database state as the meaningful check. Doing nothing then looks identical to correctly refusing the request.

Our goal is to implement a change that allows for this outcome validity issue to be resolved. Rather than just hand fixing the affected tasks as proposed in Issue 384, we want to build a general framework/detector for these broken reward specifications and measure what happens after they are repaired.

Task and Environment

We plan to work on the existing τ^3 Bench Airline and Retail environments [2]. For every episode, there is a task, user goal, and initial database state, written policy, and set of tools available to the agent. A simulated user interacts with the agent while the agent calls tools that enable it to modify/query the environment. An episode ends when either side stops, or the step limit is reached. The transcript and final environment state are then automatically scored.

Our repair focuses on cases where the task contains a requirement that is not actually enforced by reward. For instance, a task might have the agent refuse and explain why, while the evaluator only checks that the database remained unchanged. We hope to detect these mismatches automatically and repair the score so that required communication, actions, and state changes are actually included when determining success.

Metric

First, we will measure how many tasks can be passed by an agent that does nothing. Right now that is 23 of 50 airline tasks and 6 of 114 retail tasks.

After modifying the evaluator, we want trivial agents to fail while agents that actually think through the task still pass. We will test this with a null agent, which refuses to do tasks or transfers to a human, and a real model such as GPT 4.1.

We then want to compare model scores before and after the fixes and see whether the corrected evaluator changes reported performance in a meaningful way. We also want to ensure that valid solutions are not being rejected just because they use a different sequence of actions than the reference trajectory.

All scoring will remain automatic using the existing database, action, communication, and natural language assertion checks in the benchmarking infrastructure.

Baselines

Because our proposal is a fix to the existing benchmark, our main baseline is quite clear and obvious. We will compare the behavior of a null agent under our evaluator and under the original τ Bench evaluations. We'd also want to test a capable model, likely GPT 4.1, that scores well on a majority of tasks and uses the correct method to get to the final state for the questions that do give credit if you do nothing. This model should score identically with the original benchmark and in our proposed evaluator.

What will exist by October 2

By October 2, we hope to have a first draft of an automatic checker that properly detects rewards and criteria mismatches for airline retail tasks, especially those that originally give credit for doing nothing. This means that by this date, we will also have before-and-after results for the output validity scores received by a null agent and a capable agent, such as GPT 4.1.

The main risk is whether our proposed solutions will actually be successful. This has been an open issue in τ Bench for over a year since the initial publication of the Zhu et al. paper. It is possible this means that this is a non-trivial problem to solve, so it is possible that we can get stuck attempting multiple potential solutions that do inevitably fail. We do have our own potential fix ideas, and there are proposed solutions written in GitHub issue #384, but if these ideas do not work, we will need to pivot to another potential direction.

Compute and API Access

Most of this project should be cheap to run. The main cost will come from running GPT 4.1 across the airline and retail tasks for multiple trials.

References

[1] Y. Zhu et al. *Establishing Best Practices for Building Rigorous Agentic Benchmarks.* arXiv 2507.02825, 2025.

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[3] A. Cuadron et al. *SABER.* arXiv 2512.07850, 2025.

[4] Sierra Research τ² Bench GitHub Issue 384. *No op and missing reward checks in Airline and Retail tasks allow false positive rewards.* 2026.

[5] AGI Eval Official. *tau2 bench revised.* 2026.

[6] S. Yao et al. *τ Bench: A Benchmark for Tool Agent User Interaction in Real World Domains.* arXiv 2406.12045, 2024.